Case 36: Slope-factor Rothermel sweep

1 Verification purpose

This case verifies the scalar Rothermel slope response for FBFM2, FBFM9, GR2, TU5, and TL8 over 0, 10, 20, 30, 50, 75, and 100% rise over run. It tests slope-raster units, conversion to $\tan^2\phi$, fuel-dependent packing response, and ELMFIRE's slope-factor cap without wind. The verification question is whether all 35 simulation conditions reproduce independently calculated intensity and slope-enhanced rate of spread (ROS) within the stated tolerances.

2 Mathematical formulation and numerical implementation

The no-wind response is

$$R = R_0(1 + \phi_s), \qquad \phi_s = 5.275\,\beta^{-0.3}\tan^2\phi. \tag{1}$$

Here reported slope percent is rise/run times 100 and the raster value is the angle $\phi = \tan^{-1}(s/100)$ in degrees. The formulation follows Andrews (2018), RMRS-GTR-371. The inspected ELMFIRE source additionally caps ϕ_s at the maximum wind factor derived from its wind limit; the reference calculation reproduces and reports that implementation policy.

3 Derivation of expected behavior

Before the cap, plotting ϕ_s against $\tan\phi$ on logarithmic axes has slope two. For each fuel and raster angle, the local reference calculation calculates R_0, the uncapped factor, the cap, and final R. Because the initial level set is planar and aligned with the upslope direction, the output normal speed equals this scalar head-spread prediction.

The slope-only reference uses

$$\phi_s = 5.275\,\beta^{-0.3}\tan^2\theta, \qquad R = R_0(1 + \phi_s).$$

As an explicit generated substitution, the requested 20 percent slope is stored as the integer raster angle $\theta = 11$ degrees ($100\tan 11^\circ = 19.4380$ percent). For Fuel Model 2, $\beta = 0.00575$, $R_0 = 0.8237955$ m min^{-1}, and

$$\phi_s = 5.275(0.00575)^{-0.3}\tan^2(11^\circ) = 0.936756,$$
$$R_{\text{ref}} = 0.8237955(1 + 0.936756) = 1.595491 \text{ m min}^{-1}.$$

Reaction intensity is unchanged by slope; here it is 675.184 kW m^{-2}. Every other generated fuel/slope pair uses the same equations with its stored integer angle and fuel packing ratio. Simulation condition errors are $e_R = |R_{\text{max}} - R_{\text{ref}}|/\max(|R_{\text{ref}}|, 10^{-8})$ and analogously e_I. From $t = a + bx_{\text{surface}}$, $R_{\text{TOA}} = 60/b$ and $R^2 = 1 - SSE/SST$. The aggregate substitutions are $\max e_R \le 0.02$, $\max e_I \le 0.01$, $\max e_{\text{TOA}} \le 0.05$, and $\min R^2 \ge 0.995$.

4 Verification metrics and acceptance criteria

Maximum relative errors are 0.02 for direct ROS, 0.01 for reaction intensity, and 0.05 for time of arrival (TOA)-derived ROS; TOA regression requires $R^2 \ge 0.995$. The direct and reference calculation ROS

values are slope-surface speeds. Accordingly, the TOA regression converts map easting to distance along the sloping surface using $x_s = x_0 + (x - x_0)/\cos\theta$ before fitting speed. Pointwise analytical errors are used instead of a trend-only test because degrees-versus-percent mistakes can preserve monotonicity. All simulation conditions and output fields must exist.

5 Simulation configuration and predicted outcome

The finite-strip PHI raster is a signed-distance field clipped to ELMFIRE's supported $[-1, 1]$ interval, preserving both an ignited interior and a resolved front gradient. For every positive reference spread rate, the prescribed and maximum timesteps use the whole-second value obtained by rounding the 5%-of-cell target downward, with a one-second minimum; every generated value remains below the configured Courant–Friedrichs–Lewy (CFL) limit of 0.2. The stop time is rounded upward to an exact integer multiple of that step. This avoids a shortened final step that can trigger ELMFIRE's stalled-front branch and corrupt the terminal timestamp. A zero-ROS simulation condition retains a 5 s step and is evaluated from direct fields without requiring TOA propagation. The evaluator always requires uniquely named rasters tied to one final dump record. It accepts a stalled-final record only when subtracting the configured meteorology interval reconstructs the requested stop time within one configured timestep. A TOA-required simulation condition must reach the prescribed stop time in its generated namelist; a direct-only simulation condition may terminate early because lack of propagation is part of the state being tested. Each 1400 m square domain uses 10 m cells and a finite ignition strip with a planar east face in the interior analysis corridor. The strip does not touch the lateral grid boundaries. Wind is zero, dead moisture is fixed at 5/7/9%, and live moisture is 90%. Aspect is chosen so upslope is aligned with propagation. Slope inputs are generated as explicit integer degrees after applying the positive-value equivalent of Fortran NINT; the expected response and slope-surface coordinate conversion use that same generated angle. ROS should rise quadratically in $\tan\phi$ before any source-defined plateau.

Parameter	Configured value	Units	Role
Domain/grid	1400 m square; 10	m/cell	Planar upslope propagation
Fuel sweep	FBFM2, FBFM9, GR2, TU5, and TL8	code/name	Fuel-dependent packing response
Slope sweep	0, 10, 20, 30, 50, 75, and 100%	percent rise/run	Independent terrain target
Wind/moisture	calm; dead 5/7/9, live 90/90	mph / percent	Isolates slope factor
Timing	generated whole-second step and exact-multiple stop	s	CFL-safe terminal evidence

Figure 1 shows the prescribed spatial condition used in this experiment. The displayed field represents the initial level-set geometry or a controlling fuel, terrain, moisture, wind, or canopy quantity, as identified by the figure labels. The plotted values come from the prepared simulation inputs, not from the expected response.

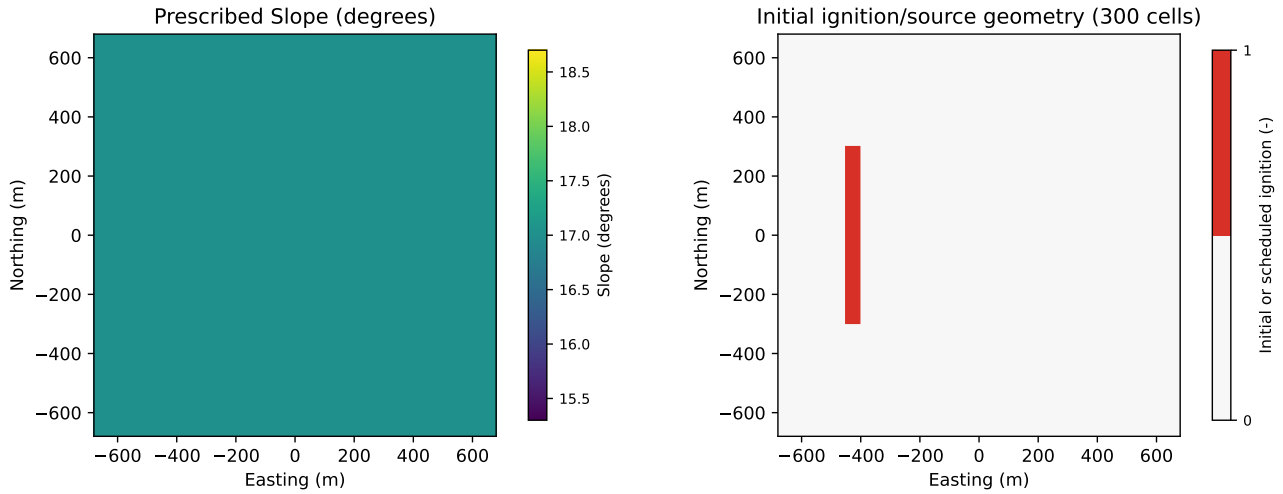


Figure 1: Prepared input conditions from the prescribed spatial fields. Coordinates are in metres; the colorbar identifies the displayed field and its physical units or category. The ignition panel is derived directly from $\phi \leq 0$.

6 Actual simulation results

The available results cover 35 of 35 simulation conditions and has state **PASS**.

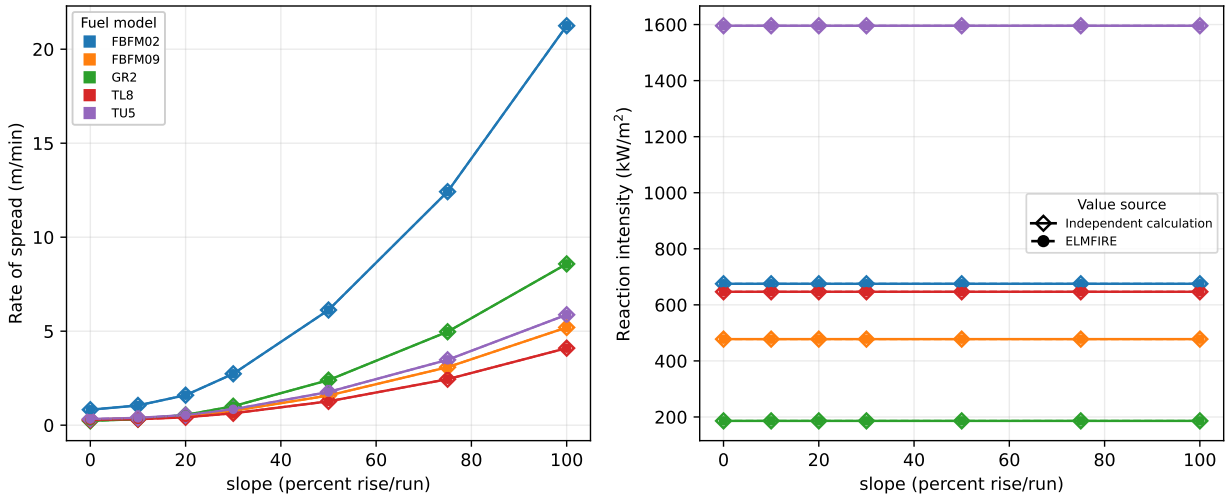


Figure 2: Independent calculations (open diamonds, solid lines) and ELMFIRE predictions (filled circles, dashed lines). Each fuel-model pair uses the same color.

7 Calculated metrics and verification decision

Metric	Limit	Calculated	Status
maximum direct-ROS relative error	≤ 0.02	5.13349e-07 fraction	PASS
maximum reaction-intensity relative error	≤ 0.01	2.93279e-07 fraction	PASS
maximum TOA-ROS relative error	≤ 0.05	0.0265981 fraction	PASS
minimum TOA regression R2	≥ 0.995	0.999961 dimensionless	PASS

The overall decision is **PASS**.

8 Technical interpretation

Large errors growing with slope indicate a units or tangent error. Fuel-dependent errors implicate packing ratio or table selection. A premature or absent plateau indicates changed cap semantics. An accurate direct field but biased TOA indicates propagation or coordinate handling.

9 Scope and limitations

This test covers uniform positive slopes aligned with the front. It does not verify vector competition between wind and slope, complex terrain, or experimental fire behavior; CASE40 covers the directional coupling.